Chapter 1 Probabilistic Reasoning

"The sun rises in the east and the rain in the west, and the road is sunny but sunny" - Liu Yuxi's "Two Bamboo Branch Words·" Part 1"

From ancient times to the present, weather prediction has been an important research topic. In ancient times, weather conditions had an important impact on agricultural and sericulture activities; In modern times, daily travel also requires attention to weather conditions. At a certain moment alone, the weather may be sunny, cloudy or rainy, and the weather conditions at a certain moment have a certain degree of randomness. Therefore, it is of great significance to accurately predict the probability of weather at a certain moment through data.

Suppose there are only three states of weather conditions in a city, namely sunny, cloudy and rainy. Without any prerequisite information, the chances of a day being sunny, cloudy, and rainy are equal. In fact, when predicting the weather, other information, such as the thickness of clouds, can be used to make predictions more accurate. With the development of technology, with the help of various sensors, a large amount of weather-related data can be collected, the data can be learned, and a model for predicting the weather can be trained.

In this chapter, we will carry out project research on the theme of "Probabilistic Statistics to Know the Weather", and understand the principles and application methods of probabilistic models in the process of predicting weather.

Theme learning project: Probability statistics to know the weather

This
chapter takes "probability statistics to know the weather" as the theme of project learning, with the help of the judgment of daily weather possibilities, learn to understand the basic knowledge of probability and statistics, understand various artificial intelligence prediction methods based on probability and statistics, understand the Monte Carlo algorithm, great likelihood estimation, Bayesian formulas and naïve Bayesian classification models lay the foundation for subsequent learning.

1. Focusing on the theme, initially master the basic knowledge of probability and statistics.
2. Learn about sampling vs. Monte Carlo algorithms from the possibility of daily weather.
3. Understanding maximum likelihood estimation allows you to infer possible weather based on a set of temperature data.
4. Understand Bayesian formulas and be able to easily apply the naïve Bayesian classification model for calculations.
5. Understand Bayesian networks and be able to apply Bayesian networks to describe causal relationships.

Project preparation
The following preparations are required to complete the project:
• The class is divided into several groups, each group is recommended to have 2~3

people, and complete the project task practice through mutual aid and cooperation in the process of learning.
• Investigate and understand the basic conditions of the local weather and prepare for subsequent learning.
• Prepare the experimental environment for the study of the topic of "Probabilistic Statistics Knowing the Weather".

Project Process
Carry out project activities while learning the content of this chapter. In order to ensure the smooth completion of the project, the progress of the project should be checked in the following stages:
(1) Learn the basic knowledge of probability.
(2) Understanding of sampling and Monte Carlo algorithms.
(3) Use great likelihood estimation to speculate on the likelihood of rain on a given day.
(4) The possibility of rain on a given day is estimated by a naïve Bayesian classification model.
(5) Estimating the influence of multiple factors on weather using Bayesian networks.

The project summarizes
and completes the series of tasks of "Probability Statistics Knowing the Weather", and each group submits learning results, carries out exchanges and evaluations, experiences the process of group cooperation, project learning and knowledge sharing, and initially masters the basic application of probability and statistics in artificial intelligence.

1.1 Probability and statistics

Learning Objectives
Understand the basic concepts of probability, conditional probability, and joint probability;
be able to describe the relationship between probability and statistics;
Master simple probability calculation methods.

Experience and explore
the possibility of rain today

One day, Mingming learned from the meteorological observatory that 25 of the 90 days in a city this summer were rainy, 46 were sunny, and 19 were cloudy. Based on this data, Ming Ming plans to build a simple model to characterize the likelihood of daily rain in the city during the summer.

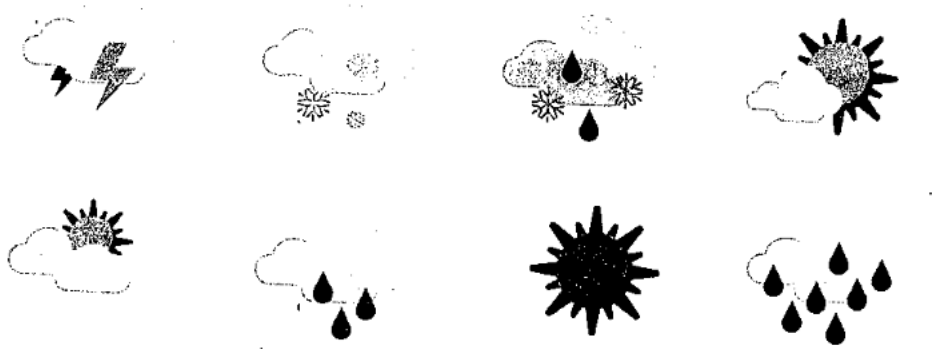


图 1-1-1 具备不确定性的天气情况

(Image-to-text expression: A set of weather icons, including lightning, clouds, snowflakes, raindrops, sun, etc., indicating the uncertainty of multiple weather conditions.)）

Figure 1-1-1 Weather conditions with uncertainty

Think

1. Is there a greater chance of sunny days or rain on any day in the city this summer?
2. How to quantitatively measure the likelihood of various weather events with the help of past weather data?

1.1.1 Probability and statistics

There are some situations in life where it is impossible to know the exact result in advance, such as the weather conditions at a certain time in the future, the traffic conditions of the road section you travel through, the difficulty of a future exam question, and the number of points after throwing a dice. The above situations are uncertain before the results appear, but the possible results have certain patterns.

In probability, these possible situations are called events. Events are composed of basic events, which are used to describe the most basic events that need to be calculated and cannot be further decomposed. The number of points that can appear on the dice roll is 1, 2, 3, 4, 5, 6, and when rolled multiple times, this 6The number of occurrences is close. Before each throw, an event may or may not occur. In probability, such events that may or may not occur in random experiments and have a certain regularity in a large number of repeated experiments are called random events. Figure 1-1-2 shows the possible outcome of the random event of the dice roll. The probability of random events occurring is called the probability of random events, and probability is often used to study the probability of various situations occurring.

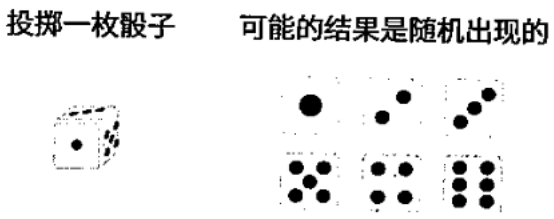


图 1－1－2　随机事件示例

(Picture to text expression: the left side is a dice, which means "throw a dice"; On the right are multiple combinations of points, indicating possible random outcomes.）

Figure 1-1-2 Example of a random event

All possible scenarios in an experiment are mutually exclusive and complete. For example, Figure 1-1-3 shows that in an experiment to throw a dice, all possible outcomes are 1, 2, 3, 4, 5, 6 any of them. After any throw, at most one of the 6 possible outcomes occurs, i.e. all possible outcomes are mutually exclusive before they occur. At the same time, at least one of the six possible outcomes occurs, that is, other outcomes are impossible, which is completeness.

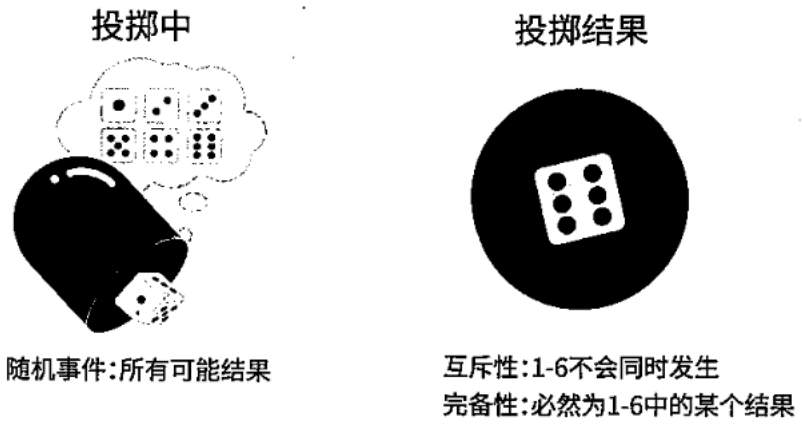


图 1－1－3　投掷骰子过程中的互斥性与完备性

(Picture-to-text expression: The picture on the left represents "throwing", including all possible outcomes; The image on the right represents the "throw result" and illustrates mutually exclusive (1–6 does not occur at the same time) and completeness (necessarily one of 1–6).）

Figure 1-1-3 Mutually exclusive and completeness in the process of throwing dice

A value is given to describe the likelihood of its occurrence, which is denoted as P(ω). The basic axiom of probability states that the probability of each possible situation in the sample space is between 0 and 1, and the total probability of all possible cases in the sample space is 1, i.e

$$0 \le P(\omega) \le 1, \sum_{\omega \in \Omega} P(\omega) = 1$$

Probabilities like P(ω) are called unconditional probabilities and usually refer to the measurement of an event without knowing other information.

How to calculate the probability corresponding to an event? Taking dice throwing as an example, assuming that the dice have a uniform texture, the probability of various points appearing in the dice throwing is 1/6, as shown in Table 1-1-1.

表 1－1－1 投掷骰子随机事件概率

	P（ω= 1）	P（ω= 2）	P（ω= 3）	P（ω= 4）	P（ω= 5）	P（ω= 6）
概率 P（ω）	1/6	1/6	1/6	1/6	1/6	1/6

Table 1-1-1 Probability of random events of dice throwing
(picture-to-text representation: the table lists ω=1 to ω=6, each corresponding probability is 1/6).

Table 1-1-1 lists all the random events that can occur in the dice throwing experiment, and their corresponding probabilities, i.e

P(ω=1)=1/6P(ω=2)=1/6
P(ω=3)=1/6P(ω=4)=1/6P(ω=5)=1/6P(ω=6)=1/6

The above can be abbreviated as:

P(Ω)=⟨1/6，1/6，1/6，1/6，1/6，1/6⟩

At this point, everything that Ω may have a predefined order (ω=1, ω=2, ω=3, ω=4, ω=5,ω=6）。 P defines the probability distribution of random events Ω, that is, the probabilities corresponding to all possible random events in an experiment.

Generally speaking, if the number of basic events that can occur in a random experiment is finite (referred to as finiteness), and the probability of each basic event occurring is the same (referred to as equal probability), a model similar to the corresponding probability of such a random experiment is called a classical probability model. In the classical probability model, if there are a basic event that constitutes event A and there are b, the probability of event A can be found to be because the probability of each basic event occurring is equal：

$$P(A) = \frac{a}{a+b}$$

Think about activities
, recognize random events and probabilities

Can you name some random events in your life? and describe how likely these random events are to occur.

The classical probability model is an idealized probability model, and whether the question belongs to the classical probability model depends on whether it has "finiteness" and "equal possibility"。 For any random experiment, sometimes it is impossible to list all the experimental results, and the experimental results may not necessarily have equal possibilities, so some random experiments are difficult to calculate the probability according to the classical probability model. So is there any other way to calculate the probability of random events instead of classical probability? Obviously there is.

Thinking about activity
dice rolling experiments

There are 3 students playing a game of dice rolling, these 3 students use a dice with a uniform texture and throw 100 times respectively, and Table 1-1-2 records the throwing results of each student.

表 1-1-2 三位同学投掷骰子 100 次的统计结果

	点数为 1	点数为 2	点数为 3	点数为 4	点数为 5	点数为 6
同学甲	17	18	15	12	18	20
同学乙	16	15	19	15	20	15
同学丙	13	15	17	16	17	22

Table 1-1-2 Statistical results of three students throwing dice 100 times
(picture to text expression: the table shows the statistics of the occurrence of points 1–6 of the three students A, B and C respectively, which are roughly close to uniform distribution).

Think according to the statistical table:

1. Compare the number of occurrences of each point to see what the pattern is?
2. How many occurrences of each point account for the total number of experiments?
3. Comparing the experimental results of the three students, what are the similarities and differences?

The ratio of the number of occurrences of a random event to the total number of experiments is called the frequency of the occurrence of the event. For example, if the dice are rolled 100 times and the number of times the number of points is 6 is 20, then for the event of rolling 6 points, its frequency is 20÷100=0.2.

Practical activities

Analyze the frequency of dice throwing experiments

According to the statistical results of the dice throwing experiment in Table 1-1-2, the frequency of each point thrown by the three students was calculated, and the throwing results of the three students were compared. Try to answer whether different students throw the same frequency for a specific point, such as point 3, and try to find patterns in comparative analysis.

In the experiment in Table 1-1-2, the frequency of throwing points of 6 by the three students is different, but the frequency values are relatively close. To be precise, the frequency at which each student throws a point of 6 oscillates around a certain constant. The number of dice casting experiments is increased to 10,000 times, and the corresponding frequency of each event is shown in Table 1-1-3.

表 1－1－3　不同实验次数对应骰子投掷结果的频率值

实验次数	点数为 1 的频率	点数为 2 的频率	点数为 3 的频率	点数为 4 的频率	点数为 5 的频率	点数为 6 的频率
100	0.1	0.14	0.22	0.18	0.16	0.2
500	0.164	0.158	0.172	0.152	0.176	0.178
1000	0.169	0.162	0.159	0.165	0.171	0.174
5000	0.1682	0.1648	0.1654	0.1672	0.1666	0.1678
10000	0.1661	0.1667	0.1668	0.1664	0.1666	0.1674

Table 1-1-3 Frequency values of dice throwing results corresponding to different experiments
(picture-to-text expression: the table lists the number of experiments as 100, 500, 1000, 5000, At 10,000, the frequency corresponding to points 1–6 gradually approaches about 0.167).

The frequency of the number of experiments is taken as the horizontal axis, and the frequency of each point is used as the vertical axis, and the law of frequency change with the number of experiments can be obtained, as shown in Fig. 1-1-4.

(Picture-to-text expression: the line graph shows that the frequency curves of different points gradually converge to the same horizontal line with the increase of the number of experiments).

Fig.1-1-4 Trend chart of frequency change with number of experiments

The change of observation frequency with the number of experiments shows that with the increase of the number of experiments, the frequency of experimental results gradually tends to a stable value. This stable value is usually considered as the probability of a random event. Therefore, the probability of a random event can be approximated using the frequency of the event in a large number of repeated experiments. According to the statistical results of a large number of repeated

experiments, the probabilities of various possible outcomes in random events can be calculated, and the probabilities obtained in this way are called empirical probabilities.

Thinking activity

Frequency and probability

Is the frequency of an event constant in multiple experiments? Is the probability of the event unchanging? What are the differences and connections between them?

Reading expansion

Pioneer of probability theory - Jacob BBernoulli

Jacob BJakob Bernoulli (1654-1705) was a Swiss mathematician, recognized as one of the pioneers of probability theory and one of the representatives of the Bernoulli family. He clarified earlier that the frequency stabilizes around probability as the number of experiments increases. He proposed the famous Bernoulli test and large number theorem in probability theory.

Jacob BBernoulli's most creative work was "Guessing", published in 1713. The Bernoulli number in this work is used in many places. The "Bernoulli theorem" he proposed was an early form of the law of large numbers.

The law of large numbers, that is, the possibility of random events occurring in a large number of repeated experiments, will present inevitable laws. To put it simply, the theorem describes that under the premise that objective conditions remain unchanged, the frequency of random events is approximately the probability of repeated experiments. For example, in the coin toss experiment, when the experiment is repeated many times, it can be found that the number of times the coin is facing up and tails is close.

It is worth mentioning that the Bernoulli family is a family of mathematicians. In addition to Jacob BIn addition to Bernoulli, during the 17th and 18th centuries, the Bernoulli family produced a total of 11 mathematicians, the more famous of which was his younger brother JohannBernoulli (1667-1748) and nephew Daniel ABernoulli (1700-1782).

1.1.2 Conditional probability

The ancients said that "the morning glow does not go out, and the evening glow travels thousands of miles", that is, the morning glow appears in the early morning, and the possibility of rain that day is very high; If there is a sunset that evening, then the chances of a sunny day are high. If the sunset appears in the evening of the same day as event B, and the sunny day on the second day is recorded as event A, then the probability that the next day will be sunny under the condition that the sunset appears in the evening of the same day is recorded as P (A| B)。
Probabilities like this are called conditional probabilities. Conditional probability usually describes the potential influence of a known factor on a possible event,

generally when the probability of event B occurring is greater than 0 (i.e., P(B)>0), the conditional event of known event B occurs The probability of A occurring is called conditional probability and is denoted as P (A| B)。

Thinking activity

Dice experiments with known partial information

Two students roll the dice, and after classmate A throws the dice, classmate B guesses the result of the throw. After a certain throw, student A told student B that the current throwing point is greater than 3.

Thinking What is the probability of a 6 thrown point $P(\omega=6 \mid \omega>3)$ under the premise that the number of points thrown is greater than 3? What is the probability of a thrown point of 1 $P(\omega=1 \mid \omega>3)$?

It is known that under the condition that the number of thrown points is greater than 3, there are only 3 possible cases in which the number of points thrown is 4, 5, and 6. Therefore, the probability of throwing a 6 point at this time is 1/3. It is known that a throw of 1 is impossible under the condition that the number of throws is greater than 3, so its probability is 0.

Without any conditions, the probability of rolling a dice with a point of 6 is 1/6; Under the condition that the number of rolls is greater than 3, the probability of rolling a dice with a score of 6 is 1/3. This shows that under certain conditions, the probability of random events may change.

Practical activities

Calculate the conditional probability distribution

If the number of points in a dice throwing experiment is known to be greater than 3, please calculate and complete the conditional probability distribution in Table 1-1-4.

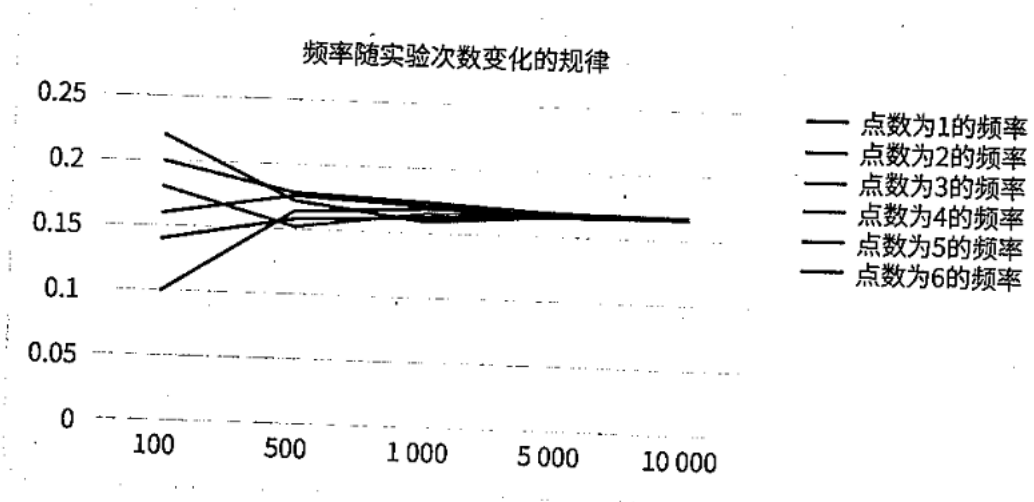


图 1－1－4　频率随实验次数变化趋势图